

Supply Chain Risk Forecasting

An Explainable AI-Based Decision Support System for Risk-Aware Inventory Management

AI for Public Good: Sustainable & Resilient Supply Chains

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Live Demo	indo-swiss-hackathon.vercel.app

2. Problem Understanding

2.1 Background

Modern supply chains operate in highly dynamic environments where customer demand is influenced by seasonal trends, promotional campaigns, pricing changes, holidays, and unexpected market disruptions. As a result, accurately forecasting product demand has become increasingly challenging.

Traditional forecasting systems typically generate a single demand estimate for each product. While these point forecasts provide an expected sales value, they offer no indication of how reliable that prediction is. Consequently, inventory planners must make replenishment decisions without understanding the uncertainty associated with the forecast.

2.2 Problem Statement

The objective of this project is to develop an intelligent inventory decision support system that not only forecasts future product demand but also quantifies the uncertainty of those predictions, explains the underlying sources of forecasting risk, and recommends appropriate inventory actions.

Rather than replacing human planners, the proposed system is designed to assist them by providing interpretable, risk-aware insights for inventory management.

2.3 Motivation

In practical inventory management, forecasting errors have significant operational and financial consequences.

- ▶ Underestimating demand can lead to stockouts, lost sales, and reduced customer satisfaction.
- ▶ Overestimating demand results in excess inventory, increased holding costs, and potential product obsolescence.

Existing forecasting approaches often fail to communicate prediction confidence, forcing managers to rely on fixed safety-stock policies or subjective judgment. This creates a need for forecasting systems that can explicitly represent uncertainty and support more informed decision-making.

2.4 Key Challenges

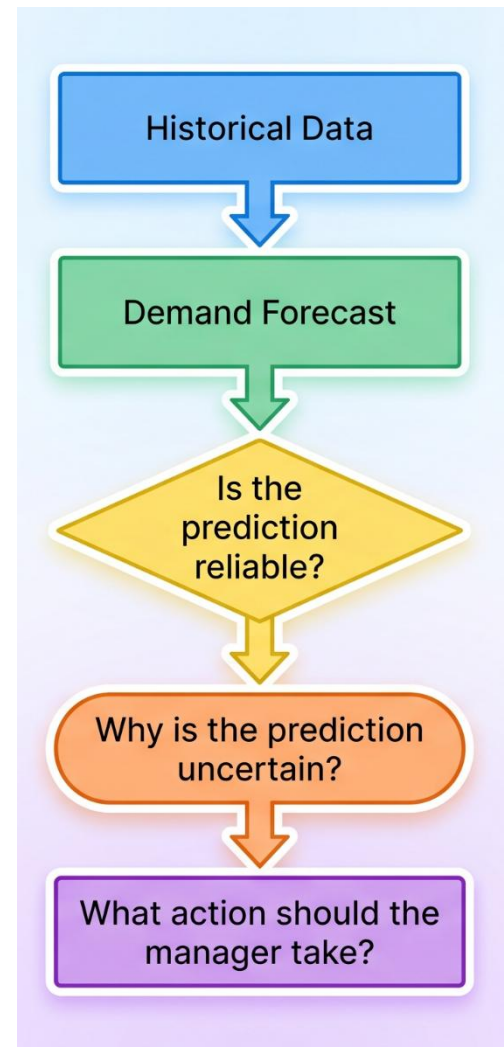
The proposed solution addresses several challenges commonly encountered in supply chain forecasting:

- Highly variable demand patterns across products.
- Intermittent demand for low-volume inventory items.
- Demand fluctuations caused by seasonal events and promotions.
- Lack of confidence estimates in conventional forecasting models.
- Limited interpretability of modern machine learning models.
- Difficulty translating model predictions into actionable business decisions.

2.5 Objectives

The primary objectives of the proposed system are:

- Develop an accurate machine learning model for demand forecasting.
- Estimate prediction uncertainty using probabilistic forecasting techniques.
- Calibrate uncertainty intervals through Conformal Prediction.
- Explain forecasting uncertainty using SHAP-based feature attribution.
- Convert model outputs into meaningful inventory risk levels.
- Recommend actionable inventory decisions through a business decision engine.
- Present results through an intuitive dashboard for supply chain managers.



3. Proposed Solution

To address the limitations of conventional demand forecasting systems, we propose an end-to-end AI-powered decision support framework for inventory risk management. Rather than generating only a point forecast, the proposed system combines probabilistic forecasting, uncertainty quantification, explainable artificial intelligence, and business decision rules to assist supply chain managers in making informed replenishment decisions.

The framework follows a multi-stage pipeline in which historical sales, pricing, calendar events, and inventory-related features are first transformed into predictive features through feature engineering. A LightGBM regression model is trained to capture complex relationships within the data and generate

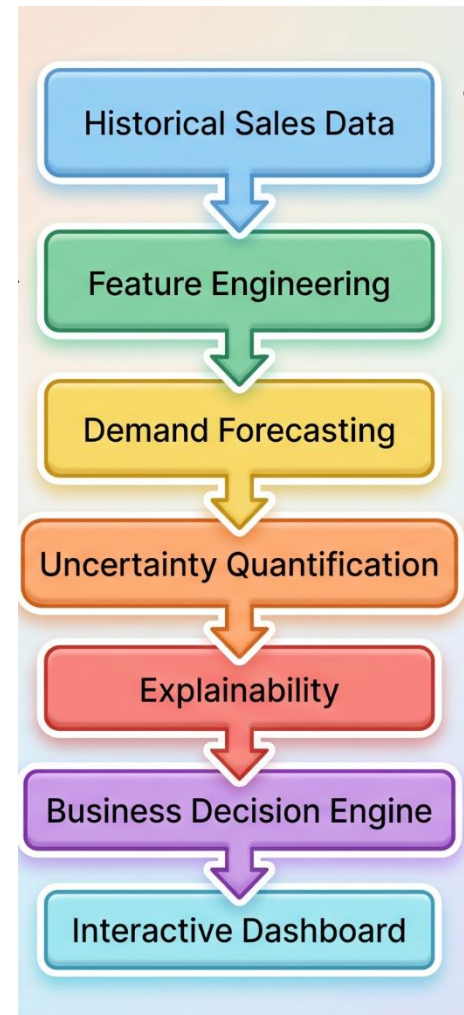
accurate demand forecasts. To quantify uncertainty, separate quantile regression models estimate the lower, median, and upper demand forecasts corresponding to the 10th, 50th, and 90th percentiles.

While quantile regression provides uncertainty estimates, these intervals may still be miscalibrated. Therefore, Split Conformal Prediction is applied to calibrate the prediction intervals, ensuring statistically valid uncertainty bounds without requiring assumptions about the underlying data distribution.

To improve transparency, TreeSHAP is used to explain model predictions by measuring the contribution of each input feature. Individual feature contributions are further aggregated into business-oriented categories such as Demand Volatility, Seasonality & Events, and Data Reliability, allowing managers to understand the primary sources of forecasting uncertainty rather than interpreting low-level model features.

Finally, the calibrated forecasts and explainability outputs are processed by a Business Decision Engine. Based on the predicted uncertainty level and dominant risk driver, the system classifies inventory risk, recommends an appropriate reorder quantity, and generates actionable inventory recommendations. These results are presented through an interactive dashboard, enabling planners to quickly identify high-risk products and make informed inventory decisions.

The proposed framework transforms traditional forecasting into a comprehensive inventory decision support system by combining prediction accuracy, calibrated uncertainty, model interpretability, and actionable business intelligence.



4. Solution Architecture

The proposed framework follows a modular, end-to-end architecture that transforms raw historical sales data into explainable, risk-aware inventory recommendations. Each module performs a well-defined task, enabling scalability, interpretability, and ease of deployment.

4.1 Data Ingestion & Feature Engineering

Objective: Transform raw sales records into predictive features suitable for machine learning.

Input Data:

- ▶ Historical daily sales
- ▶ Product identifiers
- ▶ Calendar information
- ▶ Selling prices
- ▶ SNAP (Supplemental Nutrition Assistance Program) indicators

- Promotional and event information

Feature Engineering:

The following features are generated to capture different demand characteristics:

- Lag demand features (7, 14, 28 and 56 days)
- Rolling mean statistics
- Rolling standard deviation
- Coefficient of variation (CV)
- Price lag and price change features
- Day of week, month and quarter indicators
- Time index (days since start)

4.2 Demand Forecasting Module

Objective: Predict future demand using gradient boosted decision trees.

Algorithm: LightGBM Regressor

Reason for Selection:

- Handles large datasets efficiently
- Captures complex non-linear relationships
- Supports categorical features
- Fast training and inference
- State-of-the-art performance on tabular datasets

This stage produces the baseline demand forecast used throughout the pipeline.

4.3 Probabilistic Forecasting

Objective: Estimate uncertainty rather than producing only a single demand prediction.

Three independent LightGBM Quantile Regression models are trained:

- Q10 Model → Lower demand estimate
- Q50 Model → Median demand estimate
- Q90 Model → Upper demand estimate

Instead of predicting a single value (e.g. Demand = 50), the model predicts a range:

| Q10 = 42 Q50 = 50 Q90 = 63

This allows the system to estimate the likely range of future demand.

4.4 Conformal Prediction

Objective: Calibrate prediction intervals to obtain statistically valid uncertainty estimates.

Raw quantile intervals may be overconfident or underconfident. To address this, Split Conformal Prediction is applied using a separate calibration dataset.

Procedure:

- Predict Q10 and Q90 on calibration samples.
- Compute non-conformity scores.

- Estimate the conformal adjustment factor.
- Expand prediction intervals accordingly.

Outputs:

- Calibrated Lower Bound
- Calibrated Upper Bound
- Reliable prediction intervals

This provides distribution-free uncertainty guarantees without modifying the forecasting model.

4.5 Explainability Module

Objective: Explain why a particular forecast is uncertain.

TreeSHAP is employed to compute feature-level contributions for every prediction. To improve interpretability, SHAP values are aggregated into business-oriented categories:

- Demand Trend
- Demand Volatility
- Seasonality & Events
- Data Reliability

Demand Trend is treated separately to isolate the true sources of forecasting uncertainty, resulting in a Pure Risk Breakdown that is easier for supply chain managers to interpret.

4.6 Business Decision Engine

Objective: Convert model outputs into actionable inventory decisions.

The decision engine combines: median forecast, conformal upper bound, prediction interval width, relative forecasting risk, and primary risk driver — to generate:

- Recommended reorder quantity
- Inventory risk level (Low, Medium, High, Critical)
- Primary risk driver
- Manager action recommendations

Rather than relying solely on prediction outputs, this module incorporates business rules to bridge the gap between machine learning predictions and operational decision-making.

4.7 Dashboard & Deployment

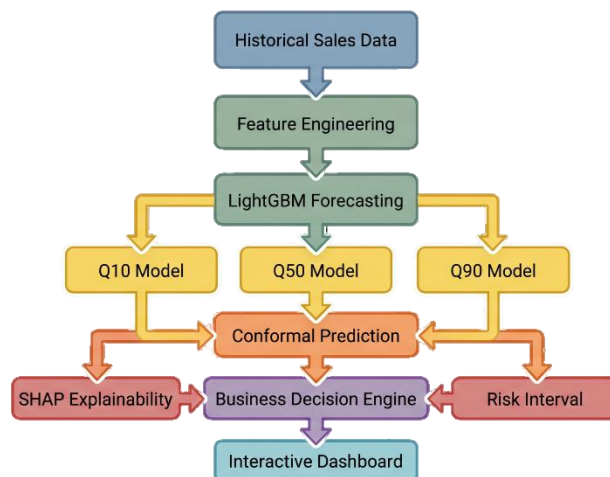
The processed outputs are exported as structured JSON records and visualized through an interactive web dashboard. The dashboard provides:

- Median demand forecast
- Safe upper demand bound
- Recommended reorder quantity
- Inventory risk level
- Pure risk decomposition
- Primary risk driver
- Manager recommendations

The frontend is implemented using ReactJS and Vite, and deployed on Vercel, enabling lightweight, browser-based access without requiring local model execution.

Architectural Characteristics:

- Modular and easily extensible
- Explainable decision-making
- Calibrated uncertainty estimates
- Business-oriented risk interpretation
- Lightweight deployment through precomputed inference
- Suitable for decision support in real-world supply chain environments



5. Technical Approach

The proposed system follows a six-stage machine learning pipeline to transform historical sales data into explainable, risk-aware inventory recommendations. Each stage is designed to progressively enhance forecasting capability, quantify uncertainty, and improve decision-making for supply chain managers.

5.1 Stage 1 – Data Preparation & Feature Engineering

The Walmart M5 Forecasting dataset was preprocessed to create a structured learning dataset suitable for supervised machine learning.

Data Sources:

- Historical daily sales
- Calendar information
- Selling prices
- SNAP indicators
- Promotional event information

Feature Engineering:

To capture temporal demand patterns, the following features were generated:

- Lag demand features (7, 14, 28 and 56 days)
- Rolling mean demand
- Rolling standard deviation
- Coefficient of variation (CV)
- Price lag
- Price change percentage
- Calendar features (day of week, week, month and quarter)
- Continuous time index

Categorical variables were encoded using LightGBM's native categorical feature handling.

5.2 Stage 2 – Baseline Demand Forecasting

A baseline LightGBM regression model was trained to estimate future product demand. LightGBM was selected because of its:

- High predictive performance on structured datasets
- Ability to model complex non-linear relationships
- Efficient training on large-scale datasets
- Native support for categorical features

The baseline model establishes the expected demand for each SKU and serves as the foundation for subsequent uncertainty estimation.

5.3 Stage 3 – Probabilistic Forecasting

Instead of generating a single prediction, three independent LightGBM Quantile Regression models were trained. The models estimate:

Model	Forecast
Q10	Lower demand estimate
Q50	Median demand estimate
Q90	Upper demand estimate

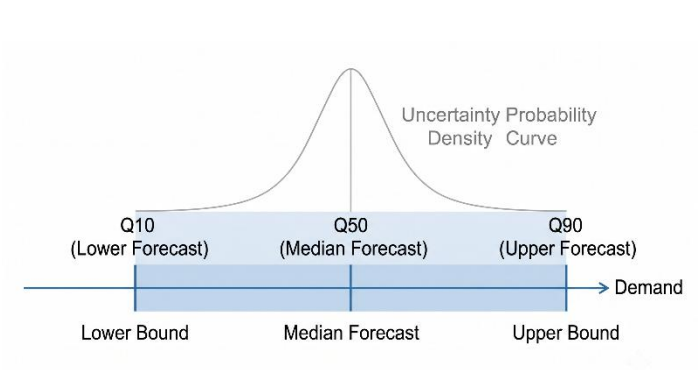


Figure 5.2. Probabilistic Forecasting using Quantile Regression

These quantile forecasts provide a probabilistic estimate of future demand, enabling the system to capture uncertainty arising from demand variability.

5.4 Stage 4 – Conformal Prediction

Although quantile regression estimates uncertainty, the resulting prediction intervals are not guaranteed to achieve the desired statistical coverage. To improve reliability, Split Conformal Prediction was applied.

Calibration Procedure:

- ▶ Reserve a calibration dataset
- ▶ Compute non-conformity scores
- ▶ Estimate the conformal adjustment factor
- ▶ Expand prediction intervals accordingly

This produces calibrated lower and upper prediction bounds with distribution-free coverage guarantees, making the uncertainty estimates more trustworthy for operational decision-making.

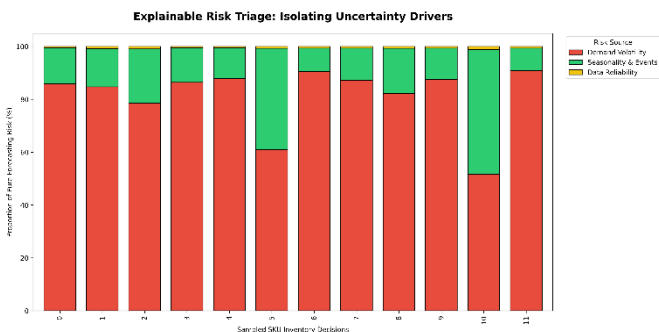
5.5 Stage 5 – Explainable AI

To improve transparency, TreeSHAP was used to explain individual model predictions. SHAP values were computed for each feature and subsequently aggregated into business-oriented categories:

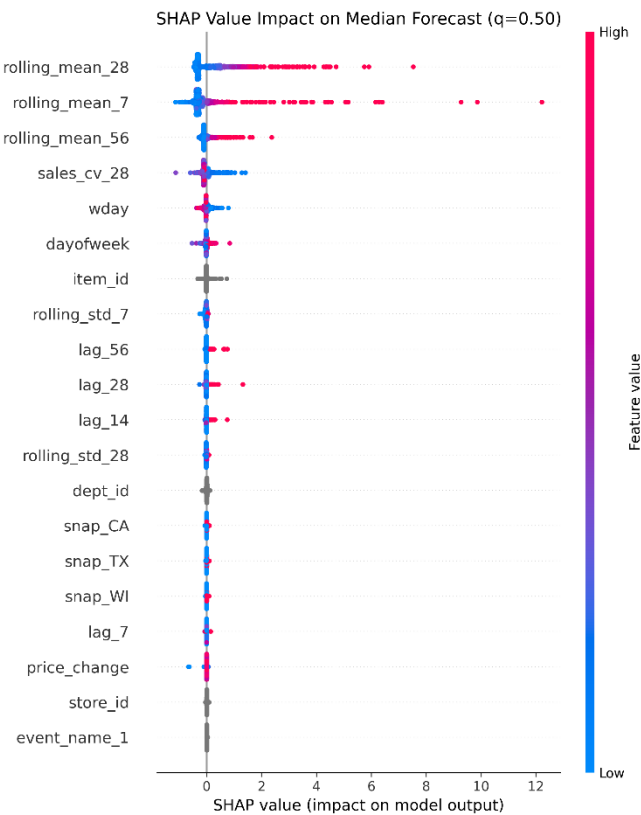
- ▶ Demand Trend
- ▶ Demand Volatility
- ▶ Seasonality & Events
- ▶ Data Reliability

Demand Trend was excluded from the pure risk decomposition because it reflects expected demand rather than uncertainty.

The resulting Pure Risk Breakdown allows managers to identify the dominant source of inventory risk rather than



interpreting dozens of individual feature contributions.



5.6 Stage 6 – Business Decision Engine

The final stage converts machine learning outputs into actionable inventory decisions. The decision engine combines:

- ▶ Median demand forecast
- ▶ Calibrated upper prediction bound
- ▶ Prediction interval width
- ▶ Relative forecasting uncertainty
- ▶ Primary risk driver

Based on these inputs, the system generates:

- ▶ Recommended reorder quantity
- ▶ Inventory risk classification (Low, Medium, High and Critical)
- ▶ Primary risk driver
- ▶ Business recommendations for inventory planners

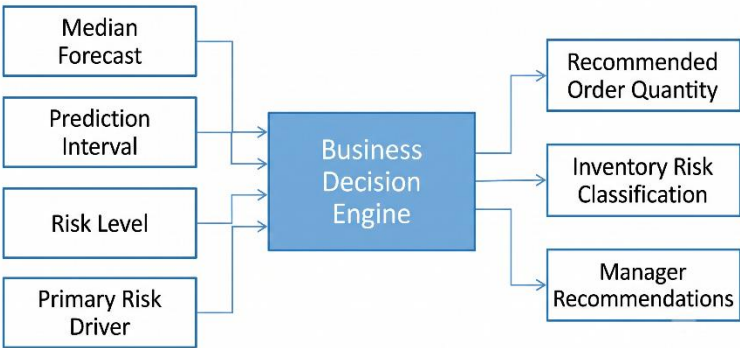


Figure 5.5: Business Decision Engine

Unlike traditional forecasting systems, the proposed framework directly links predictive analytics with operational decision support, enabling managers to make informed inventory decisions.

5.7 Prototype Implementation

The complete pipeline was implemented using Python for model development and React/Vite for the interactive dashboard.

Component	Technologies
Backend (ML)	Python, Pandas, NumPy, LightGBM, SHAP
Frontend	React, Vite, HTML5, CSS3, JavaScript

Model outputs were exported as JSON files, allowing the frontend to serve precomputed forecasts and explanations efficiently without requiring live model inference.

6. Prototype Design

The proposed solution is implemented as an interactive web-based dashboard that enables supply chain managers to visualize demand forecasts, understand forecasting uncertainty, and receive actionable inventory recommendations through an intuitive user interface.

The prototype separates machine learning from visualization. The complete forecasting pipeline is executed offline, and the generated outputs are exported as structured JSON files. The frontend consumes these files to present interactive dashboards without requiring live model inference, resulting in a lightweight and responsive application.

6.1 Dashboard Objectives

The dashboard is designed to help inventory planners:

- ▶ View predicted product demand.

- ▶ Understand forecast uncertainty.
- ▶ Identify high-risk inventory items.
- ▶ Interpret the primary causes of forecasting risk.
- ▶ Receive actionable inventory recommendations.
- ▶ Support faster and more informed replenishment decisions.

6.2 Dashboard Components

Forecast Summary

Displays the primary forecasting outputs generated by the machine learning pipeline. Information presented includes:

- ▶ Median Demand Forecast (Q50)
- ▶ Calibrated Lower Forecast (Q10)
- ▶ Calibrated Upper Forecast (Q90)
- ▶ Recommended Reorder Quantity

Risk Assessment Panel

Provides a concise summary of inventory uncertainty. The panel displays:

- ▶ Relative Risk Score
- ▶ Inventory Risk Level (Low, Medium, High or Critical)
- ▶ Prediction Interval Width

These metrics allow managers to quickly identify products requiring closer attention.

Explainability Panel

Displays the dominant contributors to forecasting uncertainty. Risk contributions are grouped into:

- ▶ Demand Volatility
- ▶ Seasonality & Events
- ▶ Data Reliability

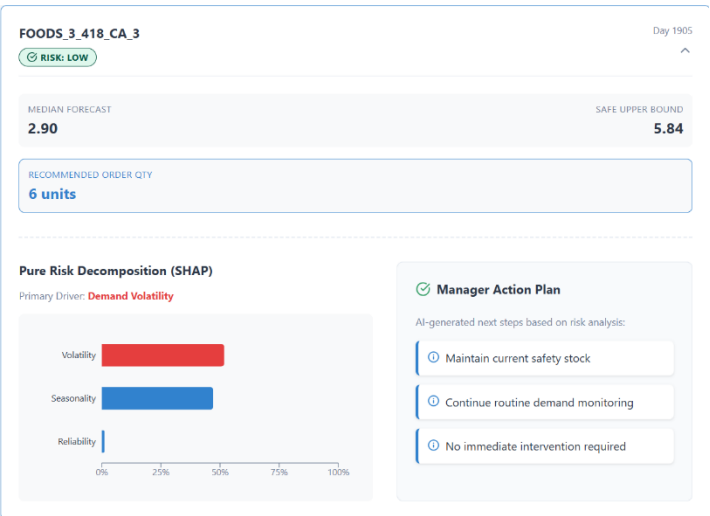
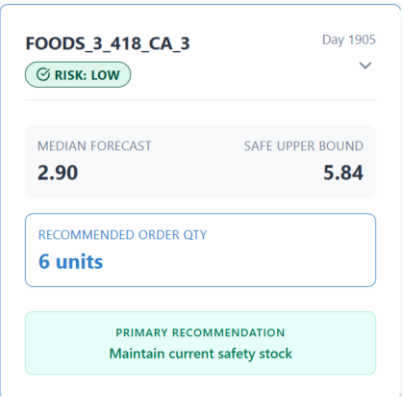
The dashboard also highlights the primary risk driver, making model predictions easier to interpret.

Recommendation Panel

The Business Decision Engine converts forecasting outputs into operational recommendations. Based on the predicted inventory risk level and dominant uncertainty source, the system recommends actions such as:

- ▶ Review safety stock levels
- ▶ Increase inventory before seasonal demand
- ▶ Coordinate with suppliers
- ▶ Validate pricing and inventory records
- ▶ Continue routine monitoring

These recommendations are intended to support—not replace—human decision-making.



6.3 Prototype Implementation

Layer	Technologies
Machine Learning Backend	Python, LightGBM, SHAP, Pandas, NumPy
Frontend	React, Vite, HTML5, CSS3, JavaScript
Deployment	Vercel (browser-based, no local installation required)

6.4 User Workflow

The typical interaction flow is:

- ▶ Select a product from the dashboard.
- ▶ View the predicted demand forecast.
- ▶ Examine the calibrated prediction interval.
- ▶ Review the assigned inventory risk level.
- ▶ Identify the primary source of forecasting uncertainty.
- ▶ Follow the inventory recommendations generated by the Business Decision Engine.

This workflow transforms complex machine learning outputs into practical decision-support information for supply chain managers.

7. Experimental Results & Evaluation

The proposed framework was evaluated on the Walmart M5 Forecasting dataset to assess forecasting accuracy, uncertainty calibration, explainability, and practical decision-support capability. The evaluation focuses on both machine learning performance and business usefulness.

7.1 Forecasting Performance

A LightGBM-based quantile regression model was trained to predict the lower (Q10), median (Q50), and upper (Q90) demand quantiles for each inventory item. To ensure reliable uncertainty estimates, Split Conformal Prediction was applied to calibrate the prediction intervals.

The final model generates:

- ▶ Median demand forecast (Q50)
- ▶ Lower prediction bound (Q10)
- ▶ Upper prediction bound (Q90)
- ▶ Recommended reorder quantity

Volatility Tier	Coverage	Mean Width	MAE
Low	87.1%	4.96	1.5
Medium	89.0%	2.46	0.82
High	89.5%	1.18	0.34

Table 7.1: Prediction interval performance across different demand volatility groups.

7.2 Uncertainty Calibration

Prediction interval quality was evaluated using coverage and interval width. The calibrated conformal intervals achieved coverage close to the desired confidence level while maintaining reasonably narrow prediction intervals.

The volatility audit further demonstrates that the uncertainty estimates remain well calibrated across products with different demand characteristics. This analysis confirms that the proposed conformal prediction framework provides consistent uncertainty estimates for both stable and highly variable demand patterns.

7.3 Explainability Analysis

Model interpretability was evaluated using TreeSHAP. Instead of presenting feature-level explanations directly, SHAP values were aggregated into business-oriented categories:

- Demand Volatility
- Seasonality & Events
- Data Reliability

This aggregation enables inventory planners to identify the dominant source of forecasting uncertainty without requiring expertise in machine learning.

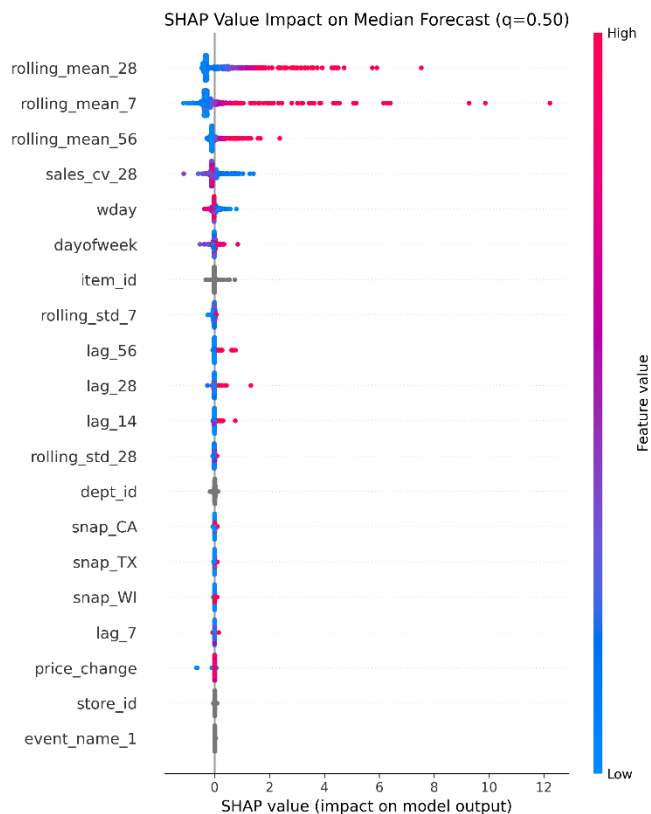


Figure 7.1: Global SHAP feature importance illustrating the contribution of individual features to demand prediction.

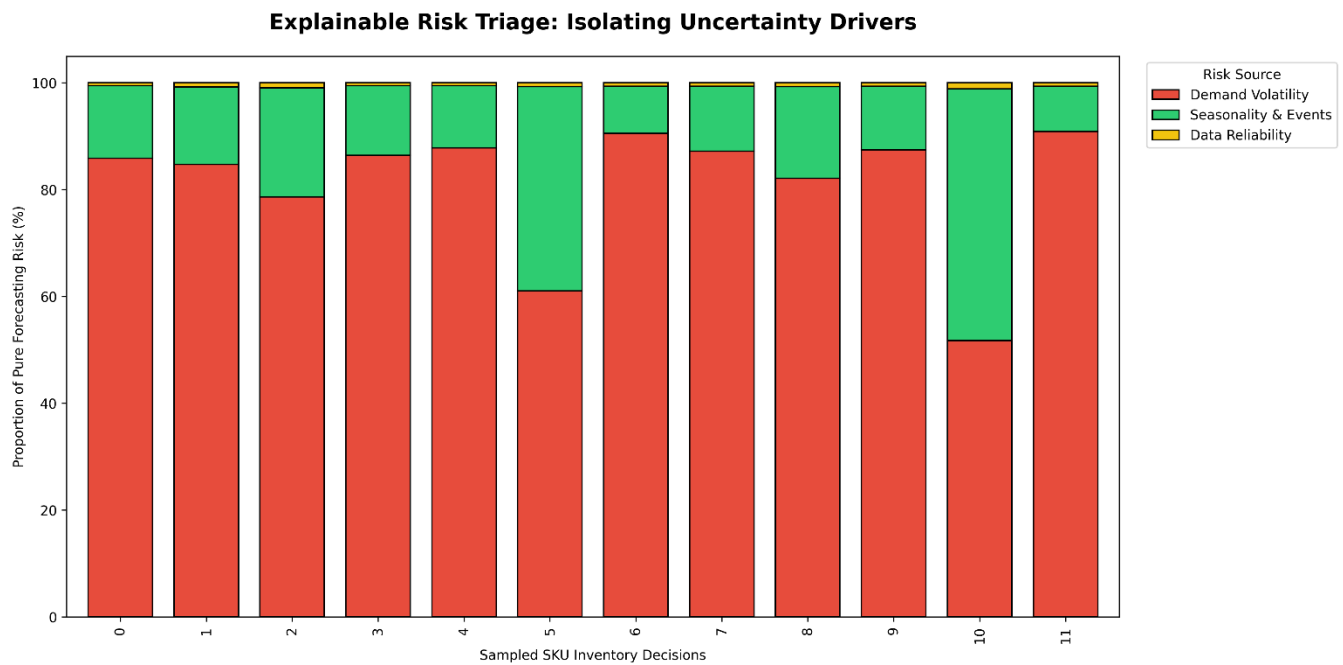


Figure 7.2: Business-oriented decomposition of forecast uncertainty into Demand Volatility, Seasonality & Events, and Data Reliability.

7.4 Business Decision Support

Beyond demand forecasting, the proposed system converts predictive outputs into operational recommendations. For every inventory item, the framework generates:

- Inventory risk classification
- Primary risk driver
- Recommended reorder quantity
- AI-generated manager recommendations

These outputs transform raw forecasting results into practical decision-support information that can be directly utilized by supply chain planners.

7.5 Prototype Evaluation

The developed prototype successfully integrates forecasting, uncertainty quantification, explainability, and decision support into a single interactive dashboard. The dashboard enables users to:

- Monitor inventory forecasts.
- Analyze prediction uncertainty.
- Understand the primary causes of forecasting risk.
- Receive AI-assisted inventory recommendations.
- Support inventory planning through an intuitive interface.

This demonstrates the feasibility of deploying the proposed framework as a practical decision-support tool for supply chain management.

8. Feasibility Analysis

The proposed framework has been designed with a strong emphasis on practical deployment, computational efficiency, and scalability. Rather than serving as a research-only prototype, the system demonstrates how explainable machine learning can be integrated into existing inventory planning workflows with minimal operational overhead.

8.1 Technical Feasibility

The complete forecasting pipeline is implemented using widely adopted open-source technologies, making the solution easy to reproduce, maintain, and extend.

Stack	Technologies
Machine Learning	Python, LightGBM, SHAP, Pandas, NumPy, Scikit-learn
Frontend	React, Vite, HTML5, CSS3, JavaScript

These technologies are lightweight, well documented, and supported by large developer communities, ensuring long-term maintainability.

8.2 Deployment Feasibility

The prototype separates model computation from visualization. The machine learning pipeline is executed offline to generate demand forecasts, prediction intervals, explainability outputs, and manager recommendations — all exported as JSON files loaded by the React dashboard.

This architecture offers several practical advantages:

- ▶ Low runtime computational cost
- ▶ Fast dashboard loading
- ▶ Simple deployment
- ▶ No GPU required during inference
- ▶ Easy integration with existing enterprise dashboards

The frontend is deployed using Vercel, enabling browser-based access without requiring software installation.

8.3 Scalability

The modular design allows the system to scale across larger retail environments. The framework can be extended to:

- ▶ Millions of inventory records
- ▶ Multiple warehouses
- ▶ Multiple retail locations
- ▶ Additional product categories
- ▶ Larger historical datasets

Since LightGBM is optimized for large tabular datasets, the forecasting pipeline remains computationally efficient even as data volume increases.

8.4 Business Feasibility

Unlike conventional forecasting systems that only generate demand estimates, the proposed framework provides explainable decision support. Business users receive forecast demand, prediction confidence, inventory risk level, primary risk driver, and AI-generated recommendations — reducing the effort required for manual forecast interpretation.

8.5 Practical Advantages

The proposed solution offers several operational benefits:

- Explainable AI for improved decision transparency.
- Calibrated uncertainty estimates for risk-aware planning.
- Actionable recommendations rather than raw predictions.
- Lightweight deployment through precomputed JSON outputs.
- Modular architecture that can integrate with existing ERP or inventory management systems.

8.6 Current Limitations

While the prototype demonstrates the feasibility of the proposed framework, several limitations remain:

- Forecasts are generated offline rather than in real time.
- The current implementation evaluates a representative subset of inventory items for visualization.
- Recommendations are generated using a rule-based Business Decision Engine built on explainable model outputs.
- External factors such as weather, supplier disruptions, and geopolitical events are not incorporated into the current forecasting pipeline.

These limitations provide opportunities for future enhancement without affecting the core functionality of the proposed system.

9. Expected Impact

The proposed framework extends traditional demand forecasting by integrating uncertainty quantification, explainable artificial intelligence, and decision support into a unified inventory planning system. Rather than providing only demand predictions, the solution equips supply chain managers with interpretable insights and actionable recommendations, enabling more informed inventory decisions.

9.1 Operational Impact

The proposed system enables inventory planners to proactively manage inventory risk by combining calibrated demand forecasts with uncertainty estimates and explainability. Key operational benefits include:

- Improved inventory planning through calibrated demand forecasts.
- Early identification of products with elevated forecasting uncertainty.
- Risk-aware replenishment decisions based on prediction intervals rather than single-point forecasts.
- Reduced dependence on manual interpretation of forecasting outputs.
- Faster prioritization of high-risk inventory items.

9.2 Business Impact

The decision-support framework has the potential to improve several key business outcomes. Expected benefits include:

- ▶ Reduced stockout risk through uncertainty-aware reorder recommendations.
- ▶ Lower inventory holding costs by avoiding excessive safety stock.
- ▶ Increased planner confidence through transparent AI explanations.
- ▶ Improved operational efficiency by prioritizing attention toward high-risk products.
- ▶ Better utilization of inventory planning resources through automated decision support.

9.3 Explainability Impact

A major contribution of the proposed system is the integration of explainable artificial intelligence into inventory forecasting. Instead of treating the forecasting model as a black box, the framework provides:

- ▶ Business-oriented risk decomposition.
- ▶ Identification of the dominant risk driver.
- ▶ Transparent explanation of forecasting uncertainty.
- ▶ Actionable recommendations linked directly to model explanations.

This improves user trust and facilitates greater adoption of AI-assisted decision-making within supply chain operations.

9.4 Scalability and Adaptability

The modular architecture enables the proposed framework to be adapted to a wide range of inventory management environments. Potential deployment scenarios include:

- ▶ Retail inventory management.
- ▶ Warehouse replenishment planning.
- ▶ Distribution center operations.
- ▶ Multi-store inventory optimization.
- ▶ E-commerce demand forecasting.

Because the architecture is modular, additional data sources and forecasting models can be incorporated with minimal modifications.

9.5 Contribution to Sustainable Supply Chains

The proposed framework contributes to sustainable and resilient supply chain management by supporting more efficient inventory utilization. By improving forecasting reliability and reducing unnecessary inventory buffers, the system can help organizations:

- ▶ Minimize inventory waste.
- ▶ Reduce excess storage requirements.
- ▶ Improve product availability.
- ▶ Enhance supply chain resilience during periods of demand uncertainty.

These objectives align with the broader goal of building intelligent, data-driven, and sustainable supply chain systems.

10. Future Scope

While the proposed framework successfully demonstrates an explainable, uncertainty-aware inventory decision support system, several enhancements can further improve its applicability in real-world supply chain environments.

10.1 Real-Time Forecasting

The current prototype performs offline forecast generation and exports the results as structured JSON files. Future versions can integrate the forecasting pipeline with real-time inventory management systems to provide:

- Live demand forecasts.
- Continuous inventory monitoring.
- Automatic risk updates.
- Real-time replenishment recommendations.

This would enable organizations to respond more rapidly to changing market conditions.

10.2 Integration with Enterprise Systems

The proposed framework can be integrated with existing enterprise software, including:

- Enterprise Resource Planning (ERP) systems.
- Warehouse Management Systems (WMS).
- Supply Chain Management (SCM) platforms.
- Inventory Management Software.

Such integration would allow forecasts and recommendations to become part of daily operational workflows.

10.3 Incorporating Additional Data Sources

Forecast accuracy can be further improved by incorporating external factors that influence demand.

Potential data sources include:

- Weather information.
- Economic indicators.
- Supplier lead times.
- Transportation delays.
- Regional events.
- Social media and market trends.

These additional signals would allow the model to capture demand fluctuations beyond historical sales patterns.

10.4 Multi-Warehouse Optimization

The current implementation focuses on forecasting demand for individual inventory items. Future work can extend the framework to support:

- Multi-warehouse inventory allocation.
- Cross-location stock balancing.
- Warehouse-specific demand forecasting.
- Regional inventory optimization.

This would enable organizations to coordinate inventory decisions across distributed supply networks.

10.5 Cost-Aware Inventory Optimization

Future versions can incorporate explicit business costs into the decision-making process. Examples include:

- Holding costs.
- Stockout penalties.
- Transportation costs.
- Supplier constraints.
- Service-level agreements.

This would allow the recommendation engine to optimize inventory decisions while balancing operational costs and customer service objectives.

10.6 Advanced Decision Support

The current Business Decision Engine uses explainable model outputs together with predefined business rules to generate recommendations. Future enhancements may include:

- Adaptive recommendation strategies based on historical planner feedback.
- Personalized recommendations for different planner roles.
- Scenario analysis under alternative demand conditions.
- Interactive 'what-if' simulations for inventory planning.

These capabilities would further strengthen the system as a comprehensive decision-support platform.

10.7 Summary

The proposed framework establishes a strong foundation for explainable, uncertainty-aware inventory forecasting. Its modular architecture allows new forecasting models, data sources, and optimization techniques to be incorporated without redesigning the overall system.

By combining probabilistic forecasting, calibrated uncertainty estimation, explainable artificial intelligence, and actionable business recommendations, the framework has the potential to evolve into a scalable decision-support platform for modern supply chain management.